

Customer Churn Analysis Using Exploratory Data Analysis (EDA)

Problem Statement - Telecom companies face significant challenges in retaining customers in a highly competitive market. Customer churn, which refers to customers discontinuing telecom services, can lead to major revenue loss and increased customer acquisition costs.

The telecom company aims to analyze customer data to identify the key factors contributing to customer churn and understand customer behavior patterns. By performing Exploratory Data Analysis (EDA), the company seeks to discover trends related to contract types, billing methods, service usage, tenure, and customer satisfaction that may influence churn.

The primary goal of this project is to help the telecom company predict potential churn risks, improve customer retention strategies, and enhance overall customer satisfaction through data-driven insights.

Objectives of the Project -

- To analyze customer behavior patterns related to churn
- To identify the major factors contributing to customer churn
- To perform data cleaning and exploratory data analysis
- To visualize relationships between customer attributes and churn
- To provide business recommendations for improving customer retention

Dataset Information -

The dataset used in this project is the IBM Telco Customer Churn dataset. It contains customer demographic information, account details, subscribed services, billing information, and churn status.

Dataset Features:

- Customer demographics
- Contract details
- Internet services
- Payment methods
- Monthly and total charges
- Customer tenure
- Churn status

Total Records: 7043

Total Columns: 21

Tools and Technologies Used-

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Project Workflow -

1. Data Collection
(Loading dataset using Pandas, importing CSV file)
2. Data Understanding
(Dataset shape, columns, data types, statistical summary)
3. Data Cleaning
(Handling null values, duplicate checking, datatype conversion, missing value treatment)
4. Feature Engineering
(Creating new features such as AvgCharges, LongTermCustomer, NumServices)
5. Univariate Analysis
(Distribution plots, countplots, histograms, single variable analysis)
6. Bivariate Analysis
(Feature vs Churn analysis, boxplots, grouped countplots)
7. Correlation Heatmap
(Correlation matrix analysis between numerical variables)
8. Outlier Detection
(Boxplots, identifying extreme values in numerical features)
9. Insight Generation
(Business insights and customer behavior analysis)
10. Conclusion and Recommendations
(Final findings, retention strategies, business recommendations)

Data Cleaning and Preprocessing -

The dataset was cleaned and preprocessed before analysis. Missing values and blank entries were handled appropriately. The TotalCharges column was converted into numeric datatype for proper analysis. Duplicate records and null values were checked to ensure data quality.

Exploratory Data Analysis (EDA) -

Univariate Analysis - Univariate analysis was performed to understand the distribution of individual variables such as gender, contract type, payment methods, monthly charges, and churn status.

Bivariate Analysis - Bivariate analysis was conducted to identify relationships between customer features and churn behavior. Various visualizations were used to compare churn across different customer groups.

Correlation Analysis- Correlation analysis was performed using a heatmap to understand relationships among numerical variables such as tenure, monthly charges, and total charges.

Outlier Detection -Boxplots were used to identify outliers in numerical features. Most outliers appeared to represent genuine high-value customers and were retained for analysis.

Key Insights-

- Customers with month-to-month contracts showed higher churn rates.
- Customers with shorter tenure were more likely to churn.
- Higher monthly charges were associated with increased churn probability.
- Customers without Tech Support and Online Security services showed higher churn behavior.
- Electronic check payment users had relatively higher churn rates.

Business Recommendations -

- Give special discounts or offers to customers who choose 1-year or 2-year contracts so they stay longer with the company.
- Pay more attention to new customers because customers with low tenure are more likely to leave the service.
- Encourage customers to use services like Tech Support and Online Security because customers using these services are less likely to churn.
- Identify customers with high monthly charges or month-to-month contracts and provide them with special offers to reduce churn.
- Improve customer support service and solve customer issues quickly to increase customer satisfaction and loyalty.

Conclusion -

The project successfully analyzed customer churn behavior using Exploratory Data Analysis techniques. The analysis identified several important factors influencing churn, including contract type, tenure, monthly charges, payment methods, and support services.

The insights generated from this analysis can help telecom companies improve customer retention strategies, reduce churn rates, and enhance overall customer satisfaction.